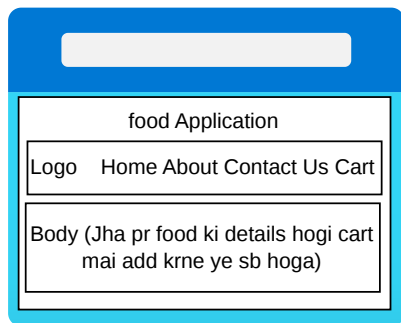


HLD (High Level Design)

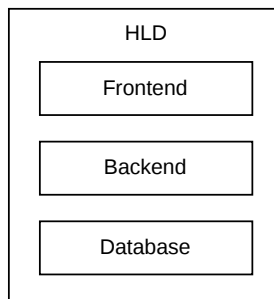
Client ki Requirement: "Mujhe 2-floor ka ghar chahiye, 3 bedrooms, 1 kitchen, 1 hall aur parking ke saath."

Master Bed Room	HLD	Room 1	HLD Mein Kya Dikh Raha Hai? - Total floors kitne hain - Rooms kaun-kaun se hain - Parking hai ya nahi - Kitchen kahan hai
Kitchen		Hall	
Parking	Gate	garden	Simple Language: HLD = Ghar ka naksha (big picture)

Technical Terms (Software Mein Iska Equivalent)



Browser



LLD (Low Level Design)

Ab HLD ghar approve ho gaya.

Engineer bolta hai:

"Ab detail mein batata hu kaise banega."

Electrical Design Bedroom [Fan] [Switch Board] [Light] [Charging Socket]	Plumbing Design Water Tank v Bathroom v Drain Pipe	Material Details Cement : UltraTech Steel : Tata Steel Tiles : Kajaria Paint : Asian Paints	Bedroom 1 Length = 12 ft Width = 10 ft Wall Thickness = 9 inch Door Width = 3 ft Height = 7 ft Window Width = 4 ft Height = 5 ft
---	--	--	---

Software Mein LLD

Maan lo HLD mein ek module tha:
User Management

LLD mein uska detail aayega:

User Class

```
User
├── userId
├── name
├── email
└── password
```

Database:

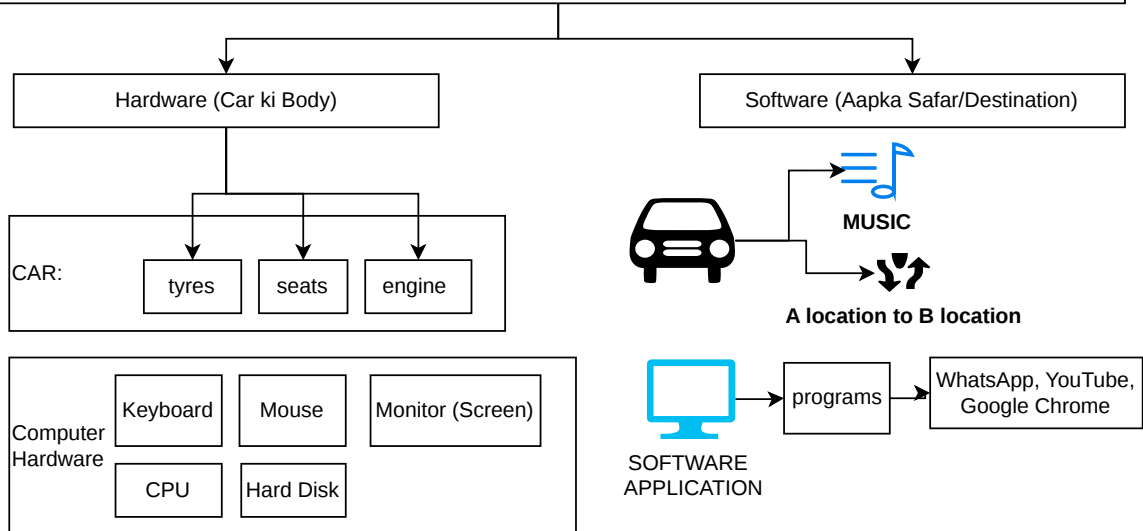
Users Table

```
user_id
name
email
password
created_at
```

API:

```
POST /register
POST /login
GET /profile
```

Computer Kya Hai?



Sabse Bada Software?

Operating System (OS) Sabse bada software

Car khadi hai, software (destination) bhi socha hua hai, par car khud se nahi chalegi. Hame chahiye ek **Driver**

Driver

Operating System

- **OPERATING SYSTEM (OS):** Yeh computer ka sabse bada software hai. Jaise hi computer chalu hota hai, OS dimaag mein aa jata hai. Bina OS ke computer sirf ek lohe ka dabba hai

Operating System ke andar ek **Kernel (Dimaag)** hota hai jo sab sambhalta hai, **Drivers** hote hain jo hardware se baat karte hain, **File System** hota hai jo photos/files ko dukan ki tarah sajata hai, aur **User Interface** hota hai jo screen par hume dikhta hai.

CPU (Central Processing Unit)

CPU computer ka asli dimaag aur engine hai. Aap jo bhi click karte hain, hisab-kitaab karte hain, ya game khelte hain, use processed (chalan) yahi CPU karta hai.

RAM (Random Access Memory)

Temporary Memory

RAM computer ki wo tez memory hai jo sirf chalu computer mein kaam karti hai. Jo app ya game aap abhi chala rahe hain, wo RAM ke upar khulta hai taaki computer bina atke tezi se kaam kare. Computer band hote hi RAM poori khali ho jaati hai.

Hard Disk / SSD (Storage)

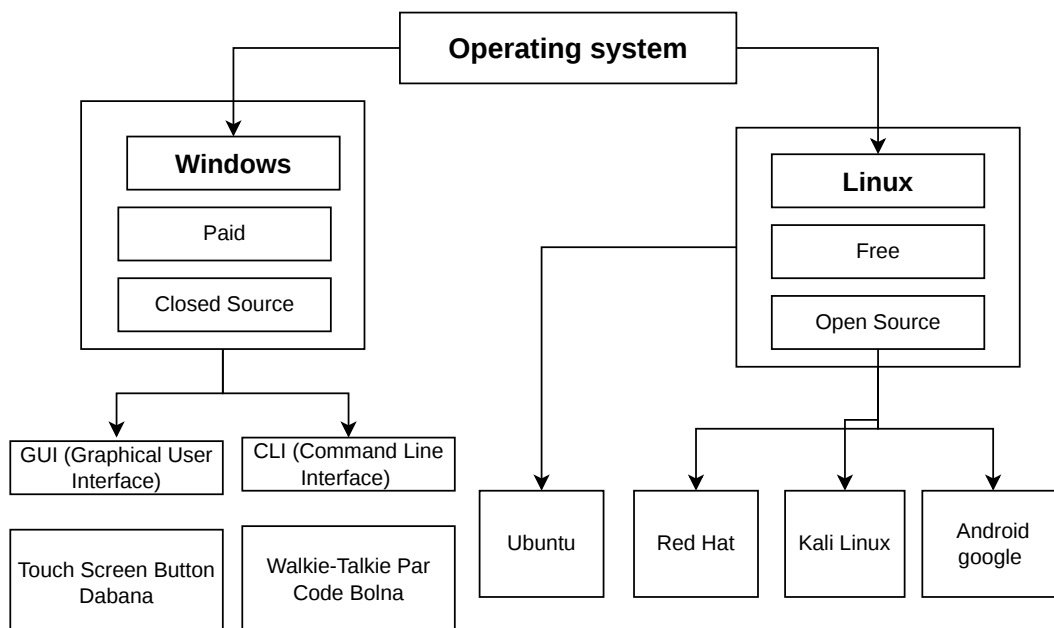
Permanent Memory

Yeh computer ka bada godam hai. Aapki saari photos, movies, software, aur gaane hamesha ke liye isme save rehte hain. Computer band hone par bhi iska data delete nahi hota. (Aajkal Hard Disk ki jagah SSD aati hai jo bohot tez hoti hai)

Graphics Card (GPU)

Video/Photo Dikhane Wala Purza

Yeh computer ka wo hissa hai jo screen par sundar dikhne wali cheezein, video games ke bhaari graphics, aur high-quality movies ko bina atke saaf-saaf dikhane ka kaam karta hai.



pwd (Print Working Directory)

ls (List)

cd (Change Directory)

mkdir (Make Directory)

touch touch diary.txt

rm (Remove) rm diary.txt